

7. What is the pH value of a salt made up of a strong acid and a weak base?

- (a) More than 7 (b) Less than 7
(c) Between 10 to 14 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

The pH value of a salt made up of a strong acid and a weak base is less than 7 because acid is strong which retains the pH value less than 7.

Acidic → Strong Acid + Weak base → pH value less than 7

Basic → Weak Acid + Strong base → pH value more than 7

Neutral → strong Acid + Strong base → pH value of 7

8. The pH of which of the following is more than 7?

- (a) Gastric juice (b) Blood plasma
(c) Vinegar (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The blood plasma is the liquid component of the blood. It's a straw-colored sticky fluid that constitutes 55% of blood's volume. The water of blood plasma functions as a solvent. Blood electrolytes such as sodium, magnesium, chloride, calcium, phosphate, potassium, and bicarbonate are inorganic salts that are found as dissolved ions in the blood plasma. Blood electrolytes function to buffer the blood and maintain its pH at approximately 7.4.

9. The gas used in the manufacture of Vanaspati Ghee from Vanaspati Oil is –

- (a) Hydrogen (b) Oxygen
(c) Nitrogen (d) Carbon dioxide

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Hydrogen gas is used in the manufacturing of vanaspati ghee. To prepare the vanaspati ghee, hydrogen gas is passed through vegetable oils under 8-10 atmospheric pressure in the presence of nickel powder (which works as catalyst) at a temperature of 150-180°C, and oil changes into vegetable ghee or solid fat known as vanaspati ghee. This process is called hydrogenation. Hydrogenation is a process in which unsaturated compounds combine with hydrogen in the presence of a catalyst and in other suitable conditions and are converted into saturated compounds.

10. The process involved in achieving Green hydrogen is–

- (a) Electrolysis (b) Dehydrogenesis
(c) Dehydration (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (a)

Green hydrogen is produced by using renewable energy to electrolyze water, which is the process of breaking down water molecules into hydrogen and oxygen using an electric current.

C. Sulphur, Nitrogen, Halogen,

Inert Gases

1. What is the shape of sulphur hexafluoride molecule?

- (a) Trigonal pyramid (b) Octahedral
(c) Planar (d) Tetrahedral

43rd B.P.S.C. (Pre) 1999

Ans. (b)

Sulphur hexafluoride (SF_6) is an inorganic, colourless, odourless and non-flammable compound. SF_6 molecule has an octahedral shape, consisting of six fluorine atoms attached to a central sulphur atom.

2. Which is used as Laughing gas –

- (a) Nitrous oxide (b) Nitrogen dioxide
(c) Nitrogen trioxide (d) Nitrogen tetra oxide

40th B.P.S.C. (Pre) 1995

Ans. (a)

Nitrous oxide (N_2O) is also known as laughing gas. It is a colourless gas with a slight metallic odour and taste. Its inhalation leads to disorientation, euphoria, numbness, loss of coordination, dizziness and ultimately a loss of consciousness. It is also used as the anesthetic gas.

3. Which of the following is also known as laughing gas?

- (a) Nitric oxide (b) Nitrous oxide
(c) Nitrogen pentoxide (d) Nitrogen
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

See the explanation of the above question.

4. Laughing gas used as anesthesia by doctors is –

- (a) Nitrogen (b) Nitrogen oxide
(c) Nitrous oxide (d) Nitrogen dioxide

44th B.P.S.C. (Pre) 2000

Ans. (c)

See the explanation of the above question.

5. Which of the following gas is used as tear gas –

- (a) H_2 (b) SO_2
(c) NH_3 (d) Cl_2

43rd B.P.S.C. (Pre) 1999

Ans. (d)

Tear gas is a chemical weapon that causes severe eye and respiratory pain, skin burning, vomiting and even blindness. In the eye, it stimulates the nerves of the lacrimal gland to produce tears. It is mostly used by the police force in controlling riots. w-chloroacetophenone o-chlorobenzylidenemalononitrile and acrolein are generally used as tear gas. Chlorine (Cl_2) is also used in preparation of tear gas.

6. Bromine is a :

- (a) Colourless gas (b) Brown solid
(c) Highly inflammable gas (d) Red liquid
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

Bromine is a member of halogen group of elements. It is a fuming red-brown liquid at room temperature that evaporates readily to form a similarly coloured gas.

7. The pH of a solution changes from 3 to 6. The H^+ ion concentration will :

- (a) increase 3 times (b) decrease 3 times
(c) decrease 10 times (d) decrease 1000 times
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

pH is a logarithmic scale used to specify the acidity or basicity (alkalinity) of an aqueous solution. It inversely indicates the concentration of hydrogen ions (H^+) in the solution as it approximates the negative of the base 10 logarithm of the molar concentration of hydrogen ions (H^+) in a solution. Each number represents a 10-fold change in the acidity/basicity of the solution. If the pH number is increased by 1, it means H^+ ion concentration is decreased by 10 times. Hence, if the pH of a solution changes from 3 to 6, the H^+ ion concentration will decrease 1000 times, and it will become less acidic.

8. Which of the following has pH value 7?

- (a) Pure Water (b) Neutral solution
(c) Basic solution (d) Acidic solution
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

The pH value of pure water and neutral solution is 7. In general, water with pH lower than 7 is considered acidic and greater than 7 is considered basic. Pure water is considered neutral.

9. Which of the following solutions has a pH value more than 7?

- (a) Buffer solution (b) Basic solution
(c) Acidic solution (d) Neutral solution
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

See the explanation of the above question.

10. Which acid is considered as a basic chemical in Industry?

- (a) H_2CO_3 (b) HNO_3
(c) H_2SO_4 (d) HCl

43rd B.P.S.C. (Pre) 1994

Ans. (c)

In the chemical industry, the sulphuric acid (H_2SO_4) is considered as basic chemical. Sulphuric acid is mainly used in petrochemical industries. It is used to refine metals, reduce corrosion of steel, refine oil and manufacture fertilizers. It is also used for cleaning purposes in the laboratory.

11. Which of the following is present in maximum amount in acid rain?

- (a) HCl (b) HNO_3
(c) H_2SO_4 (d) H_2CO_3
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

The sulphur dioxide and nitrous oxide emitted from atmospheric pollution, industrial effusions and other activities occurring in the nature reach the atmosphere and react with oxygen and cloud water and fall on earth as sulphuric acid (H_2SO_4) and nitric acid (HNO_3) respectively. This is called Acid rain. The amount of sulphuric Acid (H_2SO_4) in the acid rain is about 70-75% and it is the primary acid rain component.

12. The component(s) of acid rain is/are :

- (a) HNO_3 (b) H_2SO_4
(c) CO_2 (d) Both (a) and (b)
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

See the explanation of the above question.

13. A blackbody radiates energy at the rate of E watt per metre² at a high temperature T K. When the temperature is reduced to T/3 K, the radiant energy (in watt per metre²) will be

- (a) E /16 (b) E /27

- (c) $E/81$ (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (c)

Given that,
At temp T Kelvin, the rate of radiates energy
= E watt per square meter
As we know, $E \propto T^4$
So, at temp $t/3$, the radiant energy = $[E/3]^4$
= $E/81$

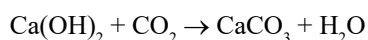
14. Limewater is turned milky by:

- (a) CO (b) CO_2
(c) O_2 (d) O_3
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Carbon dioxide (CO_2) reacts with calcium hydroxide to produce a white precipitate of calcium carbonate. Limewater is a solution of calcium hydroxide. So, if carbon dioxide is passed through limewater, the limewater will turn milky or cloudy white.



15. The compound which on dissolution in water makes it hard, is

- (a) $CaCO_3$ (b) $MgSO_4$
(c) $Ca(HCO_3)_2$ (d) More than one of the above
(e) None of the above

B.P.S.C. Headmaster 07-12-2023

Ans. (c)

The compound which dissolved in water makes the water hard is calcium bicarbonate. The Hardness of water is due to the presence of bicarbonates, sulphates and chloride of calcium and magnesium. Washing Soda, or Sodium Carbonate (Na_2CO_3), is capable of removing both temporary and permanent hardness.

16. The chemical used for destroying fungi in water tank is :

- (a) Nitric acid (b) Zinc sulphate
(c) Magnesium sulphate (d) Copper Sulphate
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

Copper sulphate (blue vitriol) is used for destroying fungi in water tank. It is also used as a fungicide to protect field crops.

17. Blue vitriol is chemically :

- (a) Sodium sulfate (b) Nickel sulfate
(c) Copper sulfate (d) Iron sulfate
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

See the explanation of the above question.

18. The chemical formula of washing soda is :

- (a) NaOH (b) Na_2CO_3
(c) $NaHCO_3$ (d) $Ca(OH)_2$

43rd B.P.S.C. (Pre) 1994

Ans. (b)

Sodium carbonate is known as washing soda. It is used as detergent. Its formula is Na_2CO_3 .

19. The liquified gas that is used in dry cleaning along with a suitable detergent is

- (a) water gas (b) CO_2
(c) petroleum gas (d) NO_2
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Liquified CO_2 (carbon dioxide) with a suitable detergent is used in dry cleaning. Tetrachloroethene ($Cl_2C=CCl_2$) was earlier used as a solvent for dry cleaning. But this solvent causes harm to the environment as it contaminates the groundwater and is also a suspected carcinogen (cancer causing agent). Thus, the process using this compound is now being replaced by a process, where liquefied carbon dioxide, with a suitable detergent, is used. Replacement of halogenated solvent by liquid CO_2 will result in less harm to groundwater. Therefore, the harmful effect will be prevented by using liquified CO_2 along with suitable detergent. The use of liquified CO_2 along with detergent will not be completely safe because most of the detergents are non-biodegradable and they cause water pollution. Moreover, liquefied CO_2 will ultimately enter the atmosphere and contribute to the greenhouse effect.

20. Quartz crystals used in clocks are :

- (a) Sodium silicate (b) Silicon dioxide
(c) Germanium oxide (d) Titanium dioxide
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

Quartz crystals used in clocks are composed of silicon and oxygen atoms. The atoms are linked in a continuous framework of SiO_4 (silicon-oxygen tetrahedron), with each oxygen being shared between two tetrahedral, giving an overall chemical formula of SiO_2 (silicon dioxide).

21. Titanium-rich mineral found in India is

- (a) bauxite
- (b) dolomite
- (c) chalcopyrite
- (d) ilmenite
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

Ilmenite and Rutile are minerals classified as heavy minerals. They are integral components of beach sand deposits found along the coastal stretches of India. Ilmenite is represented chemically as $\text{FeO} \cdot \text{TiO}_2$ and rutile as TiO_2 . Both these minerals are rich in Titanium. Titanium dioxide (TiO_2) is the key component in these minerals. These resources have significant industrial applications, particularly in the production of titanium metal, titanium dioxide pigments, and various aerospace and industrial applications.

22. Hydrofluoric acid is not kept in a glass bottle because it reacts with

- (a) Visible light
- (b) Sodium oxide of glass
- (c) Aluminum oxide of glass
- (d) Silicon dioxide of glass

46th B.P.S.C. (Pre) 2004

Ans. (d)

Hydrofluoric acid is not kept in glass. H_2F is a highly reactive compound. It reacts with the silicon dioxide in the glass, causing the bottle to explode.

23. Which among the following would cause the bright red colour due to bursting of crackers?

- (a) Strontium
- (b) Sodium
- (c) Sulphur
- (d) Magnesium
- (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

It's Because of strontium salts (strontium nitrate, strontium carbonate, strontium sulphate) red colour appears when the cracker explodes.

24. Table salt (NaCl) is a product of –

- (a) Weak acid and weak base
- (b) Strong acid and strong base
- (c) Weak acid and strong base
- (d) Strong acid and weak basic

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

The compounds formed after reaction between the strong base and strong acid are known as neutral salt. Sodium chloride (NaCl) is an example of neutral salt which is also known as Table Salt.

25. Iodised salt contains :

- (a) Free iodine
- (b) Calcium iodide
- (c) Magnesium iodide
- (d) Potassium iodide
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Potassium iodide(KI) is similar to food salt in the structure and physical properties. This is commonly used in making iodised salt. In addition to potassium iodide, sodium iodide and sodium iodate are also used in making iodised salt.

26. What is untrue for bleaching powder?

- (a) It is more soluble in water
- (b) It is powder of light yellow colour
- (c) It is an oxidant
- (d) It releases chlorine after reaction with dilute acid

39th B.P.S.C. (Pre) 1994

Ans. (a)

Calcium Hypochlorite or Calcium Oxychloride is an inorganic compound. It is also known as bleaching powder. Its chemical formula is $\text{Ca}(\text{OCl})\text{Cl}$ or $\text{Ca}(\text{OCl})_2$. It is a white solid, although commercial samples appear yellow. It is an oxidizing agent and releases chlorine after reaction with dilute acid. Bleaching powder is used for water treatment and acts as a bleaching agent. It is not highly soluble in water.

27. Human skin burns during swimming in lake/ pond -

- (a) due to infrared ray
- (b) because of chlorine
- (c) due to heat
- (d) due to ultraviolet ray

47th B.P.S.C. (Pre) 2005

Ans. (b)

Swimming in a pond burns human skin, because Cl_2 is used in pond water purification. The skin of the body gets burnt due to excess of chlorine.

28. Antiseptics and disinfectants either kill or prevent growth of micro-organisms. Which of the following statements is not true?

- (a) Chlorine and iodine are used as strong disinfectants.
- (b) Dilute solution of boric acid and hydrogen peroxide are strong antiseptics.
- (c) Disinfectants harm the living tissues.